

YOUSEF MOHAMED YOUSEF ABDELAZEM

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PROFESSIONAL SUMMARY

Second-year Artificial Intelligence student with hands-on AI engineering experience in industrial Machine Learning, Deep Learning, and Computer Vision applications. Currently working as an AI & ML Engineer at Kapci Coatings, developing confidential AI-powered systems for paint color recognition, similarity matching, and formulation prediction. Experienced in building CNN-based models, transfer learning pipelines, image preprocessing workflows, and real-world AI solutions using Python, TensorFlow/Keras, PyTorch, OpenCV, and Scikit-Learn.

EXPERIENCE

AI & ML Engineer | Kapci Coatings (Apr 2026 - Present)

- Develop confidential AI and Machine Learning solutions for industrial paint analysis, color intelligence, and formulation prediction applications.
- Build Deep Learning and Machine Learning pipelines using Python, TensorFlow/Keras, PyTorch, OpenCV, and Scikit-Learn.
- Contribute to an ongoing spectral paint formula prediction system using neural networks and spectral wavelength/reflectance data.

AI Intern | Kapci Coatings (Sep 2025 - Mar 2026)

- Developed an industrial Color Matching System achieving 97% accuracy across approximately 93 paint color classes.
- Built a CNN-based solution to recognize known colors and recommend the nearest matching color for unseen samples, supporting color engineers in reducing manual trial-and-error, material waste, and matching time.
- Developed a Metallic vs. Solid paint classification model achieving 98% accuracy using real industrial images, company website samples, and external image data.
- Improved robustness against lighting, angle, noise, and blur variations using image preprocessing and data augmentation techniques.
- Built a Python/OpenCV video processing tool to select the best frame based on lighting, blur, and noise before model inference.
- Successful delivery of the color matching project led to a full AI & ML Engineer role at Kapci Coatings.

AI Summer Trainee | Suez Canal Container Terminal - SCCT (Summer 2025)

- Completed a one-month summer training program with exposure to port operations, logistics workflows, terminal management systems, and local/international operational processes.

EDUCATION

Bachelor of Science in Artificial Intelligence

Faculty of Computers and Information, Suez Canal University | Expected Graduation: 2028

PROJECTS

Kapci Color Matching System: Confidential industrial computer vision system for paint color recognition and nearest-color matching; achieved 97% accuracy across approximately 93 classes using CNNs, transfer learning, and image preprocessing.

Metallic vs. Solid Paint Classification: Binary image classification model achieving 98% accuracy using diverse real-world and external image data; addressed lighting, angle, blur, and noise variations.

Spectral Paint Formula Prediction System - Ongoing: Confidential ML system to predict paint formula components from spectral wavelength and reflectance data using neural network-based regression models.

Selected Self-Learning ML Projects: Gold Price Prediction (regression, Kaggle data, preprocessing, evaluation); Product Success Prediction (classification, feature extraction); introductory English/Arabic OCR and digit recognition projects.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, CNNs, Transfer Learning, Regression, Classification, Feature Engineering, Data Preprocessing, Data Analysis, Model Evaluation

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib

Tools: Jupyter Notebook, Google Colab, VS Code

CERTIFICATIONS

- IBM AI Engineering Professional Certificate (13-course program)
- IBM Machine Learning Specialization (6-course program)
- Machine Learning Specialization - DeepLearning.AI & Stanford University (3-course program)
- Machine Learning Scientist in Python - DataCamp
- Associate Data Scientist in Python - DataCamp
- Python Essentials 1 & 2 - Cisco Networking Academy
- Introduction to Data Science - Cisco Networking Academy
- AI & Machine Learning Summer Camp - Microsoft & Sprints

LANGUAGES

Arabic: Native

English: Upper-Intermediate / Professional Working Proficiency